

Smoke the knives.

Do not reprint Table 1.

Balanced pack. Ablation Bot ranks 1–7. Primary smokes: planner removal, BC vs MoPA-PD, BC smoothing. Not a $3M \times 5 \times 3$ rerun.

#1	#2	#4	#3	#5	#6 α	#7
Planner off	BC vs full	Smoothing	DR	Init	Cite	Subopt.
Y	Y	Y	Partial	Partial	cite	cite

Pass: directional morals on #1 and #2 and/or #4, with motion-planner calls at inference logged as 0.
Fail: writing “we reproduced CoRL Table 1” off a Push-lite run.

Five facts that do not change with the seat

Train privilege

01

Classical MP (RRT-Connect) lives inside MoPA-RL (Yamada et al., CoRL 2020). Geometry and the planner are train-time infrastructure, not the shipped controller.

Distill, then smooth

02

Visual BC on direct-action rollouts removes the planner and smooths RRT jitter. MoPA-PD then runs asymmetric SAC: actor from BC, critic from MoPA-RL Q , $Re:R\pi = 1:3$, small α .

Deploy is planner-free pixels

03

Inference is a 32×32 image plus joints. No RRT-Connect, no object state, no motion planner. The critic's state privilege ends when training ends.

Numbers you must be able to say

04

Push 100 / 32.0 / 110.8 (ASR / AEL / ADR). Assembly 0 without BC-smooth vs 100 with. $DR \geq 96.7$ sim-to-sim (Lift Scenario 1 is worst). Expert Push AEL 111 vs Ours 32.

What this paper is not

05

Not an ancestor of ACT, Diffusion Policy, π_0 , OpenVLA, or OXE — contrast poles only. Not a JEPA / LeWM / Cosmos child. Sim-only Sawyer. Do not launder DR into sim-to-real.

What to run, what to cite, what to refuse

1	Planner-on vs distill	Y	Ours 100/99/100. MoPA-Asym 0/0/0. Push AEL 32 vs 111.
2	BC-Visual vs MoPA-PD	Y	BC 99.4/62.0/97.0. AEL ~108–118 vs 32/42/61.7.
3	Domain-rand zero-shot	Part	All ASR >96. Worst: Lift Scenario 1 = 96.7. No non-DR control.
4	BC smoothing on/off	Y	Assembly 0.0 without, 100.0 with. Fig. 8 is the jitter picture.
5	Weight init on/off	Part	≈0 until 1.2M without. ≈1.0 before 1.0M with.
6	Entropy coefficient α	Cite	Smaller $\log \alpha$ is faster. Teacher −0.63 / −2.7 / −0.29.
7	Sub-optimal expert	Cite	Push 69.2→100. Lift 41.4→99.0 after 2M visual steps.

Planner-off distill vs planner-on controls

	Sawyer Push			Sawyer Lift			Sawyer Assembly		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL	98.2	111.0	52.7	95.0	109.2	52.7	99.8	63.3	82.4
BC-Visual	99.4	118.0	46.9	62.0	108.8	34.6	97.0	115.1	50.4
Asym. SAC	0.0	250.0	0.0	0.0	250.0	0.2	0.0	250.0	3.4
MoPA-Asym. SAC	0.0	250.0	0.0	0.0	250.0	0.0	0.0	250.0	0.0
CoL	0.0	250.0	0.0	29.8	173.3	9.7	0.0	250.0	0.0
CoL w/ BC smoothing	0.0	250.0	0.0	0.0	250.0	5.1	0.0	250.0	0.0
Ours w/o BC smoothing	100.0	34.0	108.7	99.4	43.6	100.7	0.0	250.0	0.0
Ours	100.0	32.0	110.8	99.0	42.0	101.7	100.0	61.7	84.5

Table 1: Average success rate (ASR), episode length (AEL), and discounted return (ADR) of our method and baselines averaged over five seeds. Each method is evaluated after 3M environment steps. MoPA-RL [1] is the expert agent trained with state-based policy. All baselines below horizontal line are trained with asymmetric actor-critic for fair comparison. Maximum horizon is 250 for all tasks.

#1 smoke

Matched-budget Push: MoPA-PD vs Asym. SAC. MoPA-Asym only if RRT actually runs.

Hard log

Planner calls at inference = 0. Image + joints only.

Paper moral

MoPA-Asym. SAC is 0 on all three. Keeping RRT in the visual loop is not the method.

Not required

Five-seed Assembly at 100%. That number stays a citation.

BC is not the paper. Stage 2 is.

PUSH

BC-VISUAL

99.4 / 118.0 / 46.9

OURS

100 / 32.0 / 110.8

ASR already high. AEL is the gap.

LIFT

BC-VISUAL

62.0 / 108.8 / 34.6

OURS

99.0 / 42.0 / 101.7

ASR is the gap. Stage 2 is required.

ASSEMBLY

BC-VISUAL

97.0 / 115.1 / 50.4

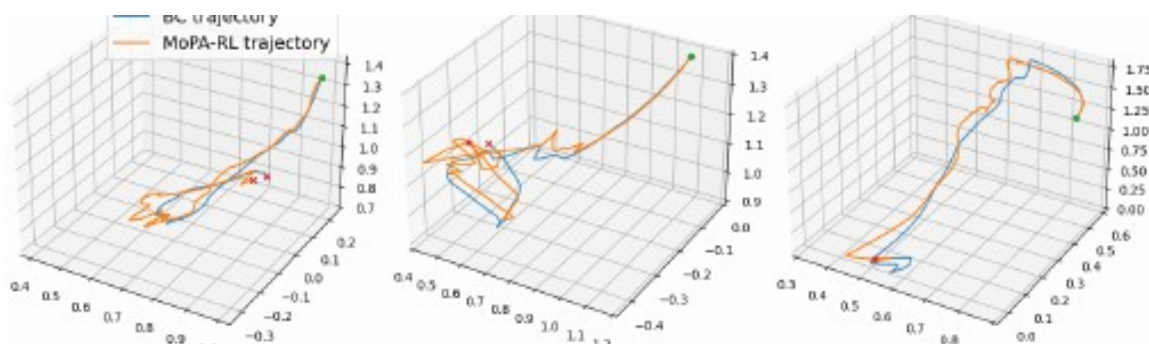
OURS

100 / 61.7 / 84.5

ASR high. Path quality is not.

Smoke one task. On Push, an ASR tie is not failure — AEL and ADR must move. Lift makes the ASR moral cleaner. Do not run all three and call it Table 1.

Smoothing is the Assembly result. Smoke Push. Cite Assembly.



Single factor

Re = BC-smoothed trajectories versus Re = raw MoPA-RL.

Assembly, paper

0 / 250 / 0

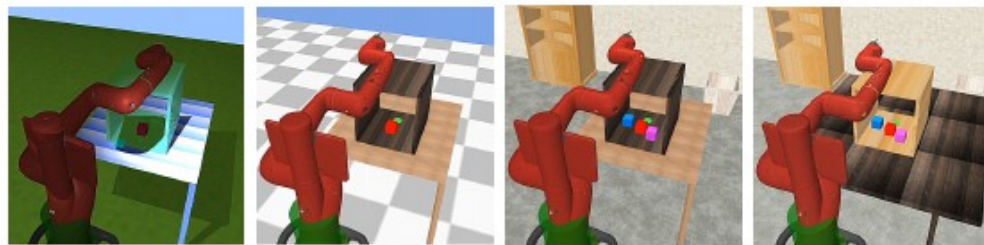
vs 100 / 61.7 / 84.5

Push, paper

Both hit ASR 100. AEL 34.0 vs 32.0. Speed and variance are the smoke signal, not a zero.

Pass on Push: smoothed Re $\geq$ raw on ASR or steps-to-threshold, and jerk or path length drops. Do not report Assembly 100% unless Assembly was run.

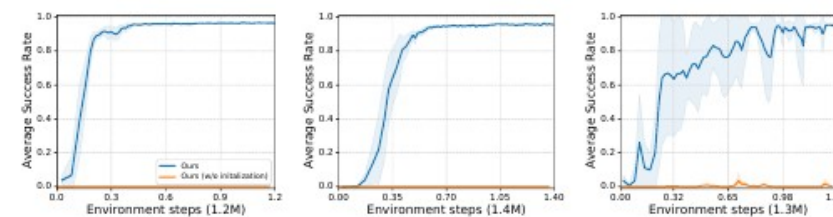
DR lite and init. Directional. Not a second paper.



(a) Domain randomization (b) Original environment (c) Scenario 1 (d) Scenario 2

#3 DR — Partial

Short DR on Push. Eval original vs a distractor stub. Paper moral: every Table 2 cell ≥ 96.7 . There is no non-DR column. Pass = OOD holds up, not “we hit 96.7.”



(a) Sawyer Push (b) Sawyer Lift (c) Sawyer Assembly

Figure 7: Learning curves comparing our method with and without actor-critic initialization for all three environments. Without initialization, the agent achieves minimal or no success rate, whereas it quickly learns to achieve optimal performance with weight initialization across all the tasks.

#5 init — Partial

Same budget, init on vs off. Prose: ~ 0 until 1.2M without; ≈ 1.0 before 1.0M with. Early-stop is fine. Log steps-to-first-success. Do not claim 1.2M unless you ran it.

α and a weak teacher. Quote the PDF. Don't grid them.

	Environment Steps					
	0.2M			1.0M		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL [1]	69.2	102.7	41.0	98.0	105.4	54.5
BC-Visual	93.2	137.4	37.1	98.8	57.4	97.2
Ours	100.0	34.4	108.2	100	32.0	110.8

(a) Sawyer Push

	Environment Steps								
	0.5M			0.52M			1.0M		
	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑	ASR ↑	AEL ↓	ADR ↑
MoPA-RL [1]	41.4	153.8	21.0	60.6	138.3	29.2	95	109.2	52.7
BC-Visual	48	189.4	14.1	55.2	146.6	24.2	63	109.4	34.8
Ours	99.0	42.0	101.6	99.4	42.9	101.0	99.0	42.0	101.7

(b) Sawyer Lift

Table 4: Average success rate (ASR), episode length (AEL), and discounted return (ADR) of our method compared with MoPA-RL and BC using sub-optimal and optimal MoPA-RL models. Even with sub-optimal training, when our state-based agent does not learn to completely solve the task, our method for distillation into a visual policy achieves high performance and successfully solves the

#6 α

Table 3, log α after MoPA-RL: Push -0.63 , Lift -2.7 , Assembly -0.29 .
Visual stage starts lower, to exploit the prior. Figs. 5–6 are qualitative. No per- α ASR table — do not make one.

#7 weak expert

Push @ 0.2M: 69.2 $\rightarrow$ Ours 100, AEL 34.4.
Lift @ 0.5M: 41.4 $\rightarrow$ Ours 99.0, AEL 42.0, after 2M visual steps.
Only if an early checkpoint already exists. Do not retrain a long teacher for #7.

Paper knobs. A smoke may shorten steps. It may not retune the story.

Image and SAC

32×32. Adam lr 1e-5. γ 0.99. Batch 256. Replay 1e6. The SAC table uses one network update per env step.

BC

lr 5e-4. Batch 512. ~1M pairs. 9:1 split. 100 expert trajectories. Select by validation success, not test MSE.

Mix and actions

Re:R π = 1:3. Action dim 7 on Push and Assembly, 8 on Lift. Reward scales 0.8 / 0.5 / 1.0. Success < 0.05. Bonus +150. Horizon 250.

What “3M” means

Teacher 1M + visual phase 2M when the teacher is required. Five seeds. 100 episodes each. That is the paper. It is not the smoke.

A course smoke is not the CoRL protocol

NEVER CLAIM TABLE 1 FROM A PUSH-LITE SMOKE

Paper scale is 3M steps $\times$ 5 seeds $\times$ 3 tasks, on top of a planner teacher. A short single-seed Push run can support a direction. It cannot support a cell.

Label runs

Put “smoke” in the filename and the first log line. Paper numbers stay in a citation column.

Frozen teacher

Prefer cached MoPA-RL demos. A from-scratch RRT teacher is out of scope.

No hardware fanfic

The PDF has BC minutes only. Do not invent GPU hours or a real-robot success rate.

Metrics

ASR without AEL and ADR hides the result. Log all three, plus planner calls at inference.

If it is not in the log, it is not a result

Identity

Task, seed, step budget, image size, frozen teacher or not.

Privilege

Planner calls at inference. Critic inputs. Init on or off.

Buffer

Re source = BC or raw. Mix 1:3 or otherwise. Expert trajectory count.

Outcome

ASR, AEL, ADR. Steps to first success. Jerk or path length. Peak VRAM.

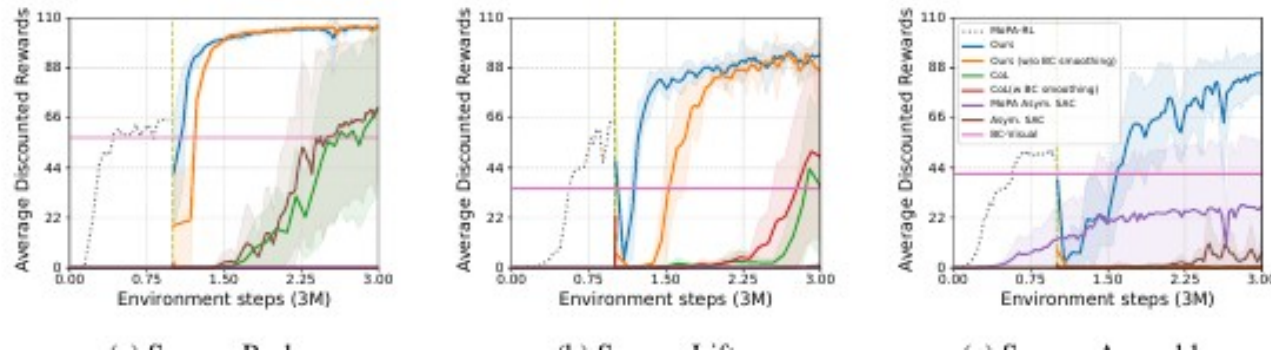
α

log α at start and end. Do not grid it until #1–#4 are logged.

DR

Train factors. Test scene id. Never the words sim-to-real.

Do not overlay a short run and call it a match



Paper curves

0 to 3M steps. Five seeds in the table. MoPA-RL takes the first 1M when it is in the method.

Your curves

Different x-axis until you have actually run 3M. Label the axis in environment steps, and label the run smoke.

What you may say

“Direction matches Fig. 3: distilled policy above asymmetric SAC.” You may not say “we reproduced Fig. 3.”

Quote these. Refuse the rest.

100 / 32.0 / 110.8. Assembly 0 without BC smoothing, 100 with. DR worst cell 96.7 — and it is still simulation.

Run

#1 planner-off, #2 BC vs full, #4 smooth vs raw. Partial #3 and #5 only if the checkpoints exist.

Do not run

3M × 5 × 3. A fresh long teacher. A real robot. A BEHAVIOR score you do not have.